

CENTRO: 180 - Escuela de Ingeniería Informática

TITULACIÓN: 4045 - Grado en Ciencia e Ingeniería de Datos

ASIGNATURA: 40386 - BIG DATA

CÓDIGO UNESCO: **TIPO:** Obligatoria **CURSO:** 3 **SEMESTRE:** 1º semestre

CRÉDITOS ECTS: 6 **Especificar créditos de cada lengua:** **ESPAÑOL:** 0 **INGLÉS:** 6

Enlace a la MEMORIA DE VERIFICACIÓN

REQUISITOS PREVIOS RESPECTO A ASIGNATURAS DE LA TITULACIÓN

Desarrollo de Aplicaciones para Ciencia de Datos
Ingeniería del Software
Fundamentos de programación

CONTENIDOS TEÓRICOS, PRÁCTICOS Y DE LABORATORIO

1. Theoretical concepts of Big Data

1.1 Complexity Management

1.2 Monitoring and Performance Engineering

1.3 Architectures for Big Data

10 horas

Kleppmann, Martin. Designing Data-Intensive Applications: The Big Ideas behind Reliable, Scalable, and Maintainable Systems. O'Reilly, 2017

2. Distributed and parallel Programming.

2.1 Parallel Programming.

2.2 Map Reduce.

2.3 Distributed file systems.

2.4 Development of applications for execution on Big Data clusters

2.5 Vector Programming.

40 horas

Kleppmann, Martin. Designing Data-Intensive Applications: The Big Ideas behind Reliable, Scalable, and Maintainable Systems. O'Reilly, 2017

3. Applications

3.1 Databases

3.2 Online data analysis

3.3 Machine learning

3.4 Graph analysis

10 horas

Kleppmann, Martin. Designing Data-Intensive Applications: The Big Ideas behind Reliable, Scalable, and Maintainable Systems. O'Reilly, 2017

Practical Content:

The practical content is directly related to the theoretical content, such that practical exercises based on the theory covered in each block will be carried out on a weekly basis.

The 2030 Agenda establishes that, in order to achieve sustainable development, action must be taken against poverty in all its forms and dimensions, inequality must be addressed, efforts must be made to preserve the planet, promote a sustainable economy, and foster social inclusion. Therefore, the commitment to sustainability must systematically address the economic, social, and environmental dimensions. The Sustainable Development Goals (SDGs) clearly present education as a key instrument for advancing sustainability.

This course adheres to the curricular sustainability guidelines issued by CRUE and ULPGC itself through the 2030 Agenda and its 17 SDGs. The competencies and content of “Big Data” will incorporate topics and references related to SDGs 5, 9, and 12; and both the teaching methodology and assessment will be guided, wherever possible, by best practices in sustainability. In particular, the syllabus and activities carried out in the course will be approached with a focus on those SDGs most closely related to Big Data.

In reviewing and correcting this section, the professor made use of artificial intelligence.

Motivos del cambio en EVALUACIÓN::

Criterios y sistemas de evaluación

Clarificar los criterios y sistemas de evaluación

Criterios de calificación

Clarificar los criterios de calificación

EVALUACIÓN:

Criterios y sistemas de evaluación

This course's assessment will be based on the following elements:

Continuous assessment model

FE1. Class Participation and Attendance

Class participation is gauged by the student's contribution to the class through asking questions, making comments, and solving exercises proposed by the instructor. Attendance is also monitored and may include responding to questions related to the topics covered during that day's session. These checks may be conducted in writing or through the course virtual learning platform.

FE2. Individual Assignments and Class Exercises

This assessment component includes the exercises and individual assignments completed throughout the course, as well as any additional activities proposed by the teaching staff. These activities provide evidence of the progressive acquisition of the knowledge and competencies addressed in each thematic block. Students will be required to present, explain, or discuss selected exercises in person during the scheduled classroom practice sessions, at the discretion of the teaching staff.

FE3. Project and Practical Activities

This assessment component includes the practical activities developed throughout the scheduled practical sessions, as well as participation in the group project. The practical activities and the group project will be formally presented and defended before the practical teaching staff. The assessment will take into account the quality and correctness of the results obtained, the clarity, rigor, and completeness of the submitted documentation, and each student's ability to explain and defend their contributions.

Non-continuous assessment model

FE2. Individual Assignments and Class Exercises

Students will complete an in-person assessment covering the theoretical and applied contents of the course. This assessment may include the resolution, presentation, explanation, or discussion of exercises and individual assignments established by the teaching staff.

FE3. Project and Practical Activities

Students must submit, present, and defend the practical activities and project established for this assessment model before the practical teaching staff.

The applicable assessment components contribute to the calculation of the final grade, with specific weightings depending on the assessment model.

The course is designed around a continuous assessment model based on regular attendance and active participation. This model considers not only the final outcomes achieved by students but also their progress throughout the semester, allowing for continuous feedback and timely adjustments to the learning process when necessary. Regular attendance is essential to this approach, as it promotes active and autonomous learning while facilitating ongoing formative assessment. This makes it possible to identify and address learning difficulties at an early stage, fostering more effective and personalized learning experiences.

In order for an assessment activity to be considered in the calculation of the final grade, it must obtain a minimum score of 5 out of 10.

Furthermore, the submission of the individual assignments, the group project, and the practical activities associated with the course is mandatory.

The final grade for the course will be expressed numerically, in accordance with Article 5 of Royal Decree 1125/2003 of September 5 (BOE of September 18), which establishes the European Credit Transfer and Accumulation System (ECTS) and the grading system applicable to official university degrees throughout Spain.

Grading System:

- 0.0 – 4.9: Fail (suspenso)
- 5.0 – 6.9: Pass (aprobado)
- 7.0 – 8.9: Merit (notable)
- 9.0 – 10.0: Distinction (sobresaliente)

If students use artificial intelligence tools in any of the activities carried out during the course, they must explicitly acknowledge such use within the corresponding submission.

Criterios de calificación

The assessment system applies to all examination periods within the academic year.

When students participate in the continuous assessment model and maintain regular attendance, the final grade will be calculated according to the following formula:

$$\text{Final Grade} = 0.1 \cdot \text{FE1} + 0.5 \cdot \text{FE2} + 0.4 \cdot \text{FE3}$$

When students do not participate in the continuous assessment model, the final grade will be calculated according to the following formula:

$$\text{Final Grade} = 0.6 \cdot \text{FE2} + 0.4 \cdot \text{FE3}$$

If a student fails to submit the mandatory individual assignments, the mandatory practical activities, or the group project corresponding to assessment components FE2 and FE3, the final record will show the grade "No Presentado" (Not Submitted).

PLANIFICACIÓN SEMANAL

Week 1: Theoretical concepts of Big Data

Theory: The course and its first topic, complexity management, are introduced. 2 hours

Practice: Students are introduced to the big data oriented group project and start the initial phase of implementation. 2 hours

Work: 6 hours

Week 2: Theoretical concepts of Big Data

Theory: Complexity management discussions continue, techniques for managing it are introduced. 2 hours

Practice: Students learn to use Git for effective version control in their big data project and continue the implementation phase. 2 hours

Work: 6 hours

Week 3: Theoretical concepts of Big Data

Theory: Solutions applying complexity management techniques are presented. 2 hours

Practice: Students learn Git Flow methodology to streamline their group project work, while continuing development. 2 hours

Work: 6 hours

Week 4: Theoretical concepts of Big Data

Theory: The second topic, performance engineering, is introduced. 2 hours

Practice: Students learn to use Maven for building and managing the big data project. 2 hours

Work: 6 hours

Week 5: Theoretical concepts of Big Data

Theory: Discussion on performance engineering issues continues, and performance engineering techniques are introduced. 2 hours

Practice: Students learn JMH for Java performance tuning in their big data project. The first iteration of the project is presented, and the next iteration is discussed. 2 hours

Work: 6 hours

Week 6: Distributed and parallel Programming

Theory: Solutions applying performance engineering techniques are presented. 2 hours

Practice: Students learn Docker for creating and managing containers in their big data project, and continue development. 2 hours

Work: 6 hours

Week 7: Distributed and parallel Programming

Theory: The third topic, parallel programming, is introduced. 2 hours

Practice: Students learn Java threading mechanisms for optimizing their big data project and continue the development phase. 2 hours

Work: 6 hours

Week 8: Distributed and parallel Programming

Theory: The discussion continues on performance issues in single-threaded applications, and parallel programming is introduced. 2 hours

Practice: Students learn Hazelcast for in-memory data grid capabilities in their big data project. 2 hours

Work: 6 hours

Week 9: Distributed and parallel Programming

Theory: Solutions applying parallel programming techniques are presented. 2 hours

Practice: Students learn Docker Compose to manage multi-container Docker applications in their big data project. 2 hours

Work: 6 hours

Week 10: Distributed and parallel Programming

Theory: The fourth topic, distributed programming (MapReduce), is introduced. 2 hours

Practice: Students learn MapReduce in Python for processing and generating big data sets in their project. The second iteration of the project is presented, and the next iteration is discussed. 2 hours

Work: 6 hours

Week 11: Distributed and parallel Programming

Theory: The discussion continues on performance issues in non-distributed applications and introduces distributed programming using MapReduce. 2 hours

Practice: Students learn MapReduce in Java with Hadoop for the processing of large data sets in their big data project. 2 hours

Work: 6 hours

Week 12: Distributed and parallel Programming

Theory: Solutions applying distributed programming techniques with MapReduce are presented. 2 hours

Practice: Students learn Nginx for load balancing in their big data project, and continue the development phase. 2 hours

Work: 6 hours

Week 13: Distributed and parallel Programming

Theory: The fifth topic, vector programming, is introduced. 2 hours

Practice: Students learn Google Cloud for deploying, scaling, and diagnosing production issues in their big data project. 2 hours

Work: 6 hours

Week 14: Distributed and parallel Programming

Theory: The discussion continues on performance issues in the experiments and vector programming using CUDA is introduced. 2 hours

Practice: Students learn to use Google Cloud with Hadoop MapReduce and distributed file usage in their big data project, and continue development. 2 hours

Work: 6 hours

Week 15: Distributed and parallel Programming

Theory: Solutions applying vector programming techniques are presented. 2 hours
Practice: Students present the third iteration of their big data oriented group project. 2 hours
Work: 6 hours

PROFESORADO

Dr./Dra. María Dolores Afonso Suárez (COORDINADOR)

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Martin Kleppmann.

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David Ríos Insúa, David Gómez-Ullate Oteiza.

Libros la Catarata :, Madrid : (2019)

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[3 Recomendado] Big data: análisis de grandes volúmenes de datos en organizaciones /

Luis Joyanes Aguilar.

Marcombo,, [Barcelona] : (2014)

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[4 Recomendado] Practical big data analyticshands-on techniques to implement enterprise analytics and machine learning using Hadoop, Spark, NoSQL and R. /

Nataraj Dasgupta.

Packt Publishing,, Birmingham : (2018)
9781783554393

[5 Recomendado] Big data architect's handbook :a guide to building proficiency in tools and systems used by leading big data experts /

Syed Muhammad Fahad Akhtar.
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